sometimes the same state-action will result in a large positive reward and sometimes in a large negative reward. If we just take the average, we will get something close to 0, which is never an observed reward in this case.

Distributional Q-learning seeks to get a more accurate picture of the distribution of observed rewards. One way to do this would be to keep a record of all the rewards observed for a given state-action pair. Of course, this would require a lot of memory, and for state spaces of high dimensionality it would be computationally impractical. This is why we must make some approximations. But first, let's delve deeper into what expected-value Q-learning is missing, and what distributional Q-learning offers.

7.1. What's wrong with Q-learning?

The expected-value type of Q-learning we're familiar with is flawed, and to illustrate this we'll consider a real-world medical example. Imagine we are a medical company, and we want to build an algorithm to predict how a patient with high blood pressure (hypertension) will respond to 4-week course of a new anti-hypertensive drug called Drug X. This will help us decide whether or not to prescribe this drug to an individual patient.

We gather a bunch of clinical data by running a randomized clinical trial in which we take a population of patients with hypertension and randomly assign them to a treatment group (those who will get the real drug) and a control group (those who will get a placebo, an inactive drug). We then record blood pressure over time while the patients in each group are taking their respective drugs. At the end we can see which patients responded to the drug and how much better they did compared to the placebo (figure 7.1).

Figure 7.1. In a randomized control trial of a drug, we study the outcome of some treatment compared to a placebo (a nonactive substance). We want to isolate the effect we are trying to treat, so we take a population with some condition and randomly sort them into two groups: a treatment group and a control group. The treatment group gets the experimental drug we are testing, and the control group gets the placebo. After some time, we can measure the outcome for both groups of patients and see if the treatment group, on average, had a better response than the placebo group.